

3.2 The A100 Datacenter GPU and Ampere Architecture

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The diversity of compute-intensive applications in modern cloud data centers has driven the explosion of GPU-accelerated cloud computing. Such applications include AI deep learning training and inference, data analytics, scientific computing, genomics, edge video analytics and 5G services, graphics rendering, and cloud gaming. The A100 GPU introduces several features targeting these workloads: a 3rd-generation Tensor Core with support for fine-grained sparsity, new BFloat16 (BF16), TensorFloat-32 (TF32), and FP64 datatypes, scale-out support with multi-instance GPU (MIG) virtualization, and scale-up support with a 3rd-generation 50Gbps NVLink I/O interface (NVLink3) and NVSwitch inter-GPU communication. As shown in Fig. 3.2.1, A100 contains 108 Streaming Multiprocessors (SMs) and 6912 CUDA cores. The SMs are fed by a 40MB L2 cache and 1.56TB/s of HBM2 memory bandwidth (BW). At 1.41GHz, A100 provides an effective peak 1248TOPS (8b integers), 624TFLOPS (FP16) and 312TFLOPS (TF32) when including sparsity optimizations. Implemented in a TSMC 7nm N7 process, the A100 die (Fig. 3.2.7) contains 54B transistors and measures 826mm².

Tensor Cores were first introduced in the Volta GPU architecture for accelerating matrix multiplies. The 3rd-generation A100 Tensor Core provides 2× higher throughput on dense FP16 matrix multiplies per SM per clock cycle, provides another 2× boost from fine-grained sparsity (Fig.3.2.2), adds comprehensive data type support, and improves energy efficiency vs. V100. Combined with a 1.35× increase in SM count, A100 supports 2.5× higher FP16 TFLOPS per GPU for dense data, and 5× for sparse data vs. V100. The A100 Tensor Cores also support new datatypes (Fig. 3.2.2). For DL training, A100 supports BF16 data and a new TF32 format for Tensor Core input operands that contains 1 sign, 8 exponent, and 10 mantissa bits. With TF32 everything outside of the tensor core remains standard FP32, including accumulators and memory storage. TF32 enables AI training to use Tensor Cores by default for FP32 data without programmer effort, providing 10× throughput on dense FP32 data and 20× for sparse FP32 data vs. V100. For DL inference, A100 supports 8b integer, 4b integer, and binary datatypes with 32b integer accumulation at even higher throughputs. For HPC, A100 introduces FP64 Tensor Cores, delivering a 2.5× FLOPS increase over V100.

The new A100 Tensor Cores consume 2× the data BW per SM vs. V100 for dense data and 3× for sparse data (compressed weights and uncompressed activations). As shown in Fig. 3.2.3, A100 adds several data movement features to keep the Tensor Cores fed, and for improved performance and energy efficiency. First, wider Tensor Cores exploit more data reuse. When 4 warps (32-thread group) worked together on a matrix multiply in V100, its 8-thread Tensor Cores would load data 4 times from shared memory into different threads' registers. In A100, the 32-thread Tensor Cores halve the shared memory load BW. Second, in V100, load-global instructions transfer data from L2 and DRAM, through the L1 cache, and into the register file (RF) before store-shared instructions write the data back to shared memory. A100 has a new combined load-global-store-shared instruction that performs an asynchronous-copy to transfer data directly into shared memory, bypassing the RF. The asynchronous copy is programmed with a new ISO C++20 asynchronous barrier [5] supported in CUDA 8.0.

In aggregate, A100's Tensor Cores run 2.5× faster than V100 for dense FP16 data but we set a design goal to not rely on weak scaling (growing the workload size by increasing the NN size or batch size) for achieving speedups. Instead, A100 targeted strong scaling with 2.5× speedup on fixed-size NNs, constraining the available parallelism and requiring more L2 cache BW per SM. To achieve this, the L2 was split into partitions using a hierarchical crossbar structure. Each L2 partition also caches data nearer to the accessing SMs, lowering latency. Hardware cache coherence still maintains the memory consistency supported by CUDA across the full GPU. A100 also adds memory system features that increase the available DRAM BW and reduce required memory traffic. An additional HBM2 site and faster clocks provide 1.56TB/s, a 1.7× increase over V100. To reduce memory traffic, A100 increases L2 capacity by almost 7× over V100 and adds L2 controls to give programmers the ability to manage the on-chip residency of data. A100 also provides compute data compression to exploit the available unstructured sparsity in NN activations, typically over 50%. A100's compression HW can compress these activation tensors in DRAM by 2-4× and in the L2 by up to 2×.

Figure 3.2.4 shows the speedups achieved per chip for A100 vs V100. For AI workloads, on MLPerf v0.7 training, A100 achieves 1.5-to-2.5× speedups vs. V100 and outperforms all other commercially available solutions. The MLPerf benchmarks also demonstrate A100's breadth of support for AI, as the only system able to run all benchmarks, and run them with high performance. The CUDA programmability and strong scaling features in A100 also benefit general-purpose datacenter workloads including HPC. A100 runs 1.5-to-2× faster than V100 on key workloads from molecular dynamics, physics, engineering, and geo science.

A100 also adds Multi-Instance GPU (MIG) features to address under-utilization of the system with small batches of work in datacenters. Each A100 can function as up to 7 isolated GPUs, reconfigurable on the fly to meet the instantaneous demand. Two types of MIG instances are supported. One type isolates compute resources, but not the memory system, enabling an OS to schedule processes with lightweight administration. The other type further provides functional and performance isolation in the memory system. In this scenario, A100 assigns each MIG physical pathways through the GPU (including the on-chip crossbar, the L2 cache, and the memory interface) that are not shared with any other MIG instances, providing flexible security boundaries for cloud computing providers.

To further support strong scaling of A100 in multi-GPU systems, I/O BW was also increased. A100's NVLink3 doubles the BW per GPU to 300GB/s in each direction through a combination of faster signaling and increasing the links per GPU to 12. NVLink3 is a long-reach (LR) differential interface operating at a raw bitrate of 50Gbps (Fig. 3.2.5). Unlike the prevailing 50Gbps LR standards which use PAM4 signaling and require the use of FEC (forward error correction) to achieve low BER, the NVLink3 PHY uses NRZ signaling and achieves 1e-15 BER without requiring a FEC. This saves tens of nanoseconds of round-trip latency. In the LR mode, the NVLink3 PHY uses a combination of 3-tap Tx FIR filter, Rx Continuous-Time Linear Equalizer (CTLE) and Partial Response Maximum-Likelihood (PRML) sequence detection to equalize the channel and boost the SNR. The Tx FIR filter is trained through a back-channel supported by the NVLink protocol. The CTLE has equalization tunability at multiple frequency bands to closely track the channel response. To relax the equalization requirement, partial response target is used. The inter-symbol correlation from a partial response target can be exploited by using sequence detection instead of symbol-by-symbol detection to boost the SNR. The Viterbi convolutional decoder is used to achieve maximum-likelihood sequence detection. A baud-rate clock recovery architecture is used to reduce the power and area of the receiver.

Figure 3.2.6 shows a DGX A100 system with 8 A100 GPUs and 6 NVSwitch chips. Each GPU connects to each of the NVSwitch chips using 2 NVLink3 links for a total of 12 NVLink3 connections supporting 600GB/s of NVLink interconnect for each A100. The system supports full-BW non-blocking communication pairwise between all the GPUs. It also contains multiple high speed Mellanox HDR 200Gb/s Infiniband NICs for connecting to massively parallel multi-GPU systems via a full fat tree switched interconnect.

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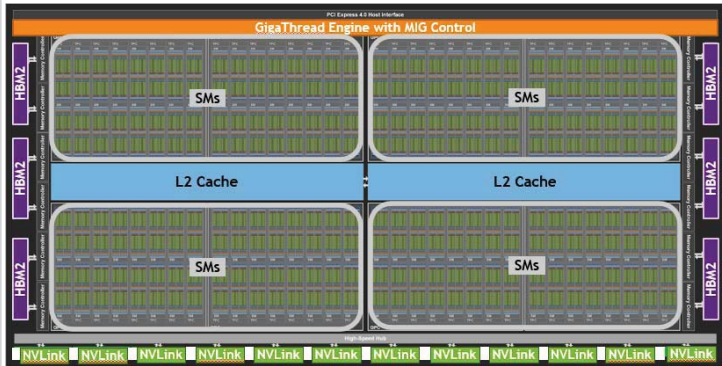


Figure 3.2.1: A100 architecture block diagram: 108 SMs (6912 CUDA cores), 40MB L2 (6.7x capacity vs. V100), 1.56TB/s HBM2 (1.7x BW vs. V100).

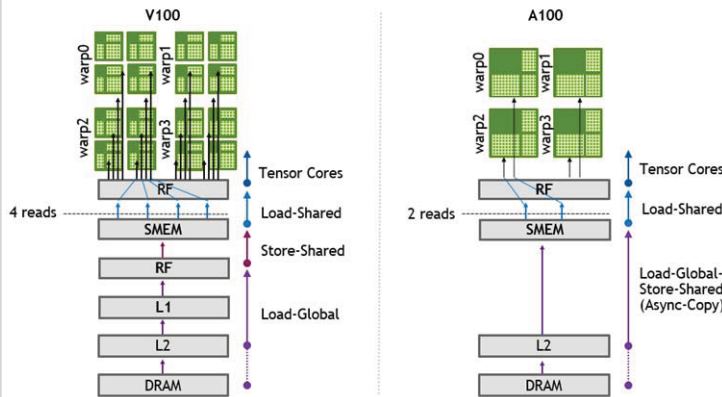


Figure 3.2.3: A100 Streaming Multiprocessor (SM) architecture and data movement efficiency for strong scaling: A100 improves SM BW efficiency compared to V100 with a new load-global-store-shared asynchronous copy instruction that bypasses L1 cache and register file (RF). Additionally, A100's more efficient Tensor Cores reduce shared memory (SMEM) loads.

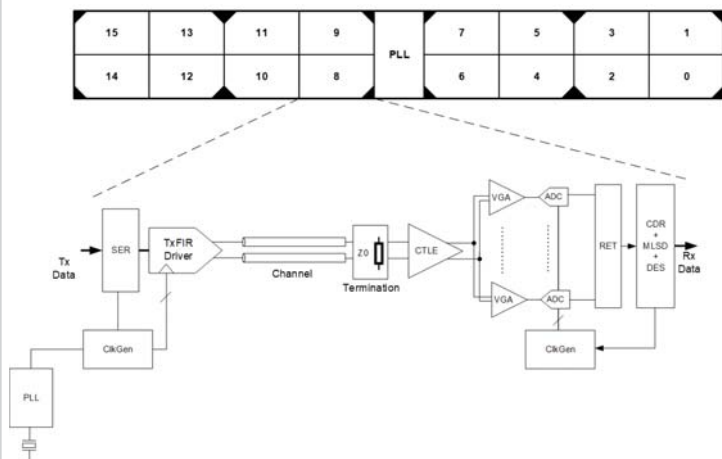


Figure 3.2.5: NVLink3 link architecture.

	Input Operands	Accumulator	TOPS	SPARSE TOPS
V100	FP32	FP32	15.7	-
	FP16	FP32	125	-
	FP32	FP32	19.5	-
	TF32	FP32	156	312
A100	FP16	FP32	312	624
	BF16	FP32	312	624
	INT8	INT32	624	1248
	INT4	INT32	1248	2496
	BINARY	INT32	4992	-
	FP64	FP64	19.5	-
	FP64	FP64	19.5	-

Figure 3.2.2: A100 Tensor Core input/output formats and performance: TOPS column indicates TFLOPS for floating-point ops and TOPS for integer ops. Sparse TOPS represents effective TOPS / TFLOPS using the new sparsity feature.

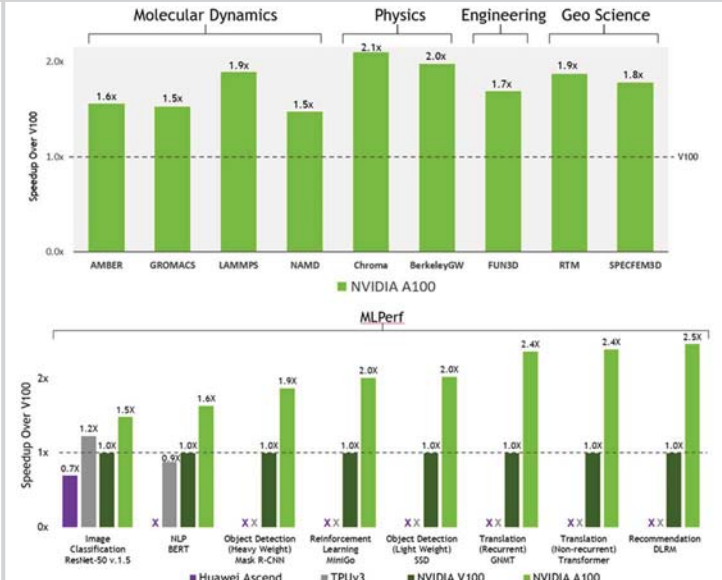


Figure 3.2.4: A100 speedups on HPC and MLPerf v0.7 vs. V100 and alternatives.

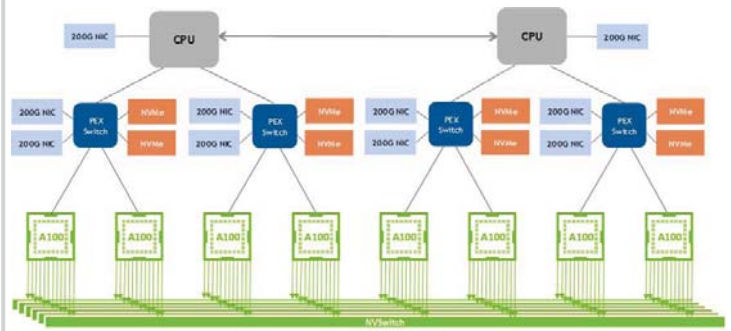


Figure 3.2.6: DGX-A100 system block diagram.

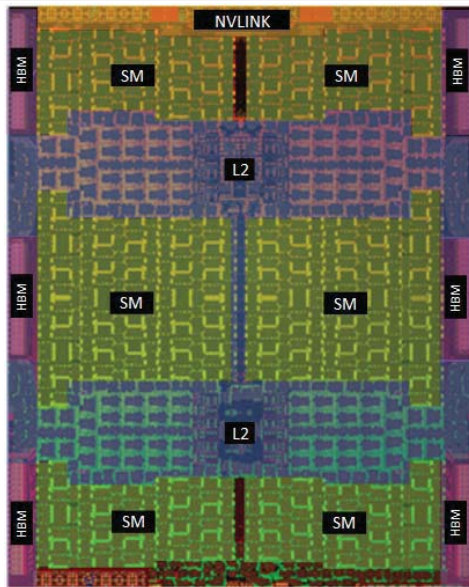


Figure 3.2.7: A100 die photo.